

Electrochemistry

NCERT Class XII Chemistry · Chapter 2 · Electrochemistry

Electrochemistry — Complete Notes

14 marks — second heaviest on the blueprint. Two engines run this chapter: the Nernst equation and Faraday's laws. Nearly every numerical is one of those two wearing a different hat.

The two cell types — *the distinction every question hangs off*

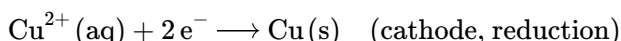
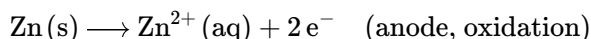
	Electrochemical (galvanic)	Electrolytic
Does	Chemical → electrical	Electrical → chemical
Reaction	Spontaneous	Non-spontaneous
ΔG	Negative	Positive
Anode	Negative	Positive
Cathode	Positive	Negative

Constant across both: **oxidation at the anode, reduction at the cathode**. Only the charge signs flip.

Examiner asks: "two points of difference" (2020) is a standing 2-marker. Give the ΔG and the anode-sign pair — they're the two that can't be waffled.

Daniell cell and cell notation — *how to write a cell down, and why the salt bridge is there*

Zinc rod in 1 M ZnSO_4 , copper rod in 1 M CuSO_4 , joined by a salt bridge and an external wire. EMF = **1.1 V** at unit concentration.



The zinc rod thins; the copper rod thickens.

Cell notation

Anode on the left, cathode on the right. Single line = phase boundary, double line = salt bridge:



What the salt bridge actually does

It's a U-tube of electrolyte (KCl, KNO₃, NH₄Cl) set in agar or gelatin. Without it, Zn²⁺ builds up in the anode compartment and SO₄²⁻ in the cathode compartment; the charge separation stalls the cell. The bridge's ions migrate in to neutralise both halves and complete the circuit.

Examiner asks: "why is a salt bridge necessary" (2026) — answer with both jobs: maintains electrical neutrality *and* completes the circuit.

Electrode potential and SHE — *why everything is written as reduction potential*

Dip a metal in a solution of its own ions and charge separates until equilibrium. That potential difference between metal and solution is the **electrode potential**.

By IUPAC convention **every electrode potential is quoted as a reduction potential** — because you can't subtract an oxidation potential from a reduction potential meaningfully. Oxidation and reduction potentials of the same electrode are numerically equal and opposite in sign.

Standard electrode potential (E°): ion concentration 1 M, 298 K, and 1 bar for any gas.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

Also written $E_{\text{right}} - E_{\text{left}}$, since cathode sits on the right in cell notation. E_{cell} for a working cell is always positive.

Standard hydrogen electrode

A single half-cell's absolute potential can't be measured — you need a second electrode to get a reading at all. So SHE is defined as the reference with $E^\circ = 0.00$ V exactly. It's a platinum foil coated with platinum black, H₂ at 1 bar, H⁺ at 1 M.

Connect zinc to SHE: electrons flow zinc → SHE, so zinc is the anode, reading 0.76 V. Then $0.76 = 0 - E^\circ_{\text{Zn}^{2+}/\text{Zn}}$, giving $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76$ V. Copper against SHE reads 0.34 V with electrons flowing the other way, giving +0.34 V.

Electrochemical series — *reading a table of E° values*

Arrange standard electrode potentials in order and the series tells you three things at a glance.

- **High E°** → reduces easily → **strong oxidising agent**. Fluorine, +2.87 V, is the strongest.
- **Low (negative) E°** → oxidises easily → **strong reducing agent**. Lithium, −3.05 V, is the strongest.
- **Feasibility:** the species actually being reduced must have the higher E° . If a proposed reaction has it backwards, the reaction isn't feasible.

Worked · 2023 — E° of a cell

$E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80$ V, $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44$ V. Find E°_{cell} .

Higher E° becomes the cathode, so silver is cathode, iron is anode.

$$E_{\text{cell}}^{\circ} = 0.80 - (-0.44) = 1.24 \text{ V}$$

Examiner asks: "which of A and B liberates H_2 from dilute H_2SO_4 " (2026) — the one with the more negative E° , because it oxidises more readily.

Nernst equation — *the workhorse* — for when concentrations aren't 1 M

Standard potentials assume 1 M. Change the concentration and the potential changes. Nernst gives the relation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q$$

At 298 K, with $R = 8.314$ and $F = 96500$, that collapses to the form you'll actually use:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{products}]}{[\text{reactants}]}$$

n = electrons exchanged. **Solids and pure liquids are omitted** — their concentration is constant, taken as unity.

Trap: n is the electrons exchanged *after balancing*, not the charge on one ion. And stoichiometric coefficients become powers inside the log. Both are where marks vanish.

Worked · 2019 — EMF with unequal concentrations

$\text{Al} \mid \text{Al}^{3+} (0.001 \text{ M}) \parallel \text{Ni}^{2+} (0.1 \text{ M}) \mid \text{Ni}$. $E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.25 \text{ V}$, $E_{\text{Al}^{3+}/\text{Al}}^{\circ} = -1.66 \text{ V}$.

Balance: Al loses 3e^- , Ni^{2+} gains $2\text{e}^- \rightarrow$ multiply by 2 and 3 $\rightarrow n = 6$.

$$E_{\text{cell}}^{\circ} = -0.25 - (-1.66) = 1.41 \text{ V}$$

$$E_{\text{cell}} = 1.41 - \frac{0.0591}{6} \log \frac{(10^{-3})^2}{(10^{-1})^3} = 1.41 - \frac{0.0591}{6} \log(10^{-3})$$

$$= 1.41 + \frac{0.0591 \times 3}{6} = 1.41 + 0.0295 = 1.44 \text{ V}$$

Nernst at equilibrium $\rightarrow$ equilibrium constant

At equilibrium $E_{\text{cell}} = 0$ and $Q = K_c$, so:

$$E_{\text{cell}}^{\circ} = \frac{0.0591}{n} \log K_c$$

Examiner asks: "why is K_c related to E_{cell}° and not E_{cell} ?" (2026) — because at equilibrium E_{cell} is zero and generates no current, while E° stays constant.

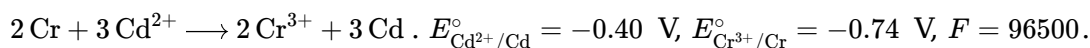
Gibbs energy and cell EMF — *one equation, two exam uses*

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^\circ = -nFE_{\text{cell}}^\circ$$

The electrical work a cell does equals the fall in Gibbs energy.

The intensive/extensive point the examiner likes: E_{cell} is **intensive** — multiply the equation by 2 and it doesn't change. ΔG is **extensive** — it depends on n , so doubling the equation doubles it.

Worked · 2025 — ΔG° and $\log K_c$



Cd is reduced $\rightarrow$ cathode. $E_{\text{cell}}^\circ = -0.40 - (-0.74) = 0.34\text{ V}$. Cr loses $3e^- \times 2 = 6$, so $n = 6$.

$$\Delta_r G^\circ = -6 \times 96500 \times 0.34 = -196860\text{ J mol}^{-1} = -196.86\text{ kJ mol}^{-1}$$

Conductance, conductivity, molar conductivity — *four quantities that are easy to confuse*

Quantity	Symbol	Relation	Unit
Resistance	R	$R = \rho \frac{l}{A}$	Ω
Resistivity	ρ	$\rho = R \frac{A}{l}$	$\Omega\text{ cm}$
Conductance	G	$G = 1/R$	S (siemens)
Conductivity	κ	$\kappa = 1/\rho$	S cm^{-1}
Molar conductivity	Λ_m	$\Lambda_m = \frac{\kappa \times 1000}{M}$	$\text{S cm}^2\text{ mol}^{-1}$

Cell constant

l/A is fixed for a given conductivity cell, so it's called the **cell constant** G^* :

$$G^* = \frac{l}{A} = \kappa \times R$$

Measuring l and A directly is unreliable, so G^* is found by filling the cell with a solution of known conductivity (usually KCl) and measuring resistance. Once known, the same cell gives κ for any solution: $\kappa = G^*/R$.

Resistance is measured on a Wheatstone bridge with an AC source (DC would electrolyse the solution and change its composition) and a purpose-built conductivity cell (a solution can't be wired into the bridge directly).

Worked · 2024 — resistivity, conductivity, molar conductivity

0.05 M NaOH, cell constant 50 cm^{-1} , resistance $4.5 \times 10^3\ \Omega$.

$$\kappa = \frac{G^*}{R} = \frac{50}{4.5 \times 10^3} = 0.011\text{ S cm}^{-1}$$

$$\rho = \frac{1}{\kappa} = 90 \, \Omega \, \text{cm}$$

$$\Lambda_m = \frac{0.011 \times 1000}{0.05} = 220 \, \text{S cm}^2 \text{mol}^{-1}$$

Variation with dilution — the two go opposite ways

- **Conductivity falls** on dilution: fewer ions per cm^3 .
- **Molar conductivity rises** on dilution: the volume holding one mole of electrolyte increases, and interionic attraction weakens.

Trap: these two moving in opposite directions is itself an exam question. Conductivity is per unit *volume*; molar conductivity is per *mole*.

Strong vs weak electrolyte on the graph

Plot Λ_m against $\sqrt{c}$:

- **Strong electrolyte** — starts high, rises gently and **linearly**. Extrapolate to zero concentration to read Λ_m° straight off the graph.
- **Weak electrolyte** — starts low, then shoots up steeply near zero concentration, running almost **parallel to the y-axis**. Cannot be extrapolated; you must use Kohlrausch's law instead.

Kohlrausch's law — *and its three applications*

Kohlrausch law of independent migration of ions: the limiting molar conductivity of an electrolyte is the sum of the individual contributions of its cation and anion.

$$\Lambda_m^\circ(\text{NaCl}) = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{Cl}^-}^\circ$$

$$\Lambda_m^\circ(\text{BaCl}_2) = \lambda_{\text{Ba}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$$

Trap: the stoichiometric multiplier. $\text{Al}_2(\text{SO}_4)_3$ needs $2\lambda_{\text{Al}^{3+}}^\circ + 3\lambda_{\text{SO}_4^{2-}}^\circ$. Forgetting the 2 and 3 is the whole mistake.

1 • Λ° of a weak electrolyte

Can't be read off a graph, so build it from strong electrolytes that share ions:

$$\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{CH}_3\text{COONa}) + \Lambda^\circ(\text{HCl}) - \Lambda^\circ(\text{NaCl})$$

The Na^+ and Cl^- terms cancel, leaving exactly $\lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{H}^+}^\circ$.

2 · Degree of dissociation

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

3 · Dissociation constant

$$K_a = \frac{c \alpha^2}{1 - \alpha}$$

Worked · 2020 — limiting ionic conductivity from Kohlrausch

$\Lambda^\circ(\text{Al}_2(\text{SO}_4)_3) = 858$, $\lambda_{\text{SO}_4^{2-}}^\circ = 160 \text{ S cm}^2 \text{ mol}^{-1}$. Find $\lambda_{\text{Al}^{3+}}^\circ$.

$$858 = 2\lambda_{\text{Al}^{3+}}^\circ + 3(160) \Rightarrow 2\lambda_{\text{Al}^{3+}}^\circ = 858 - 480 = 378$$

$$\lambda_{\text{Al}^{3+}}^\circ = 189 \text{ S cm}^2 \text{ mol}^{-1}$$

Faraday's laws of electrolysis — *the second engine of this chapter*

First law

Mass deposited is proportional to the charge passed:

$$w = \frac{M I t}{n F}$$

M = molar mass, I = current in amperes, t = time in **seconds**, n = electrons gained per ion, $F = 96500 \text{ C mol}^{-1}$.

Second law

Same charge through different electrolytes deposits masses in the ratio of their **equivalent weights**:

$$\frac{w_1}{w_2} = \frac{E_1}{E_2}, \quad E = \frac{\text{atomic mass}}{n}$$

So Na (23/1), Mg (24/2 = 12), Al (27/3 = 9) deposit in the ratio 23 : 12 : 9 per faraday.

Trap: time in minutes. Convert to seconds first — this single slip has cost more marks in this chapter than any conceptual error.

Worked · 2017 — mass deposited

2 A through AgNO_3 for 15 min. $M(\text{Ag}) = 108$.

$t = 15 \times 60 = 900 \text{ s}$. $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$, so $n = 1$.

$$w = \frac{108 \times 2 \times 900}{1 \times 96500} = 2.014 \text{ g}$$

Charge in faradays — without the formula

Often faster to reason directly. To reduce 1 mol Zn^{2+} you need 2 mol of electrons = **2 F** = 193000 C. To reduce 1 mol MnO_4^- to Mn^{2+} , Mn goes $+7 \rightarrow +2$, so 5 mol electrons = **5 F**.

Products of electrolysis — *the E° comparison, and the overpotential exception*

In aqueous solution, water competes with the dissolved ions at both electrodes. Two rules decide the winner:

- **At the cathode:** the species with the **higher** E° is reduced.
- **At the anode:** the species with the **lower** E° is oxidised.

Aqueous NaCl — the standard case

Cathode: Na^+ (−2.71 V) vs H^+ (0.00 V). Hydrogen wins $\rightarrow \text{H}_2$ gas.

Anode: Cl^- (1.36 V) vs water $\rightarrow \text{O}_2$ (1.23 V). By the rule oxygen should win — **but it doesn't**. Oxygen evolution is kinetically slow and needs extra voltage (**overpotential**) to proceed at a useful rate, so chlorine is released instead.

Net: $\text{NaCl(aq)} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{Cl}_2$

Trap: the overpotential exception is the entire point of this question. Answering "oxygen, because $1.23 < 1.36$ " is the trap being set. Name overpotential explicitly.

Molten vs aqueous

Molten NaCl has no water, so there's no competition: sodium at the cathode, chlorine at the anode.

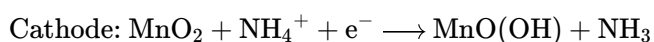
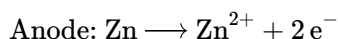
Sulphuric acid

Dilute $\text{H}_2\text{SO}_4 \rightarrow$ oxygen at the anode. Concentrated $\rightarrow$ peroxodisulphate,
 $2\text{SO}_4^{2-} \rightarrow \text{S}_2\text{O}_8^{2-} + 2\text{e}^-$ ($E^\circ = 1.96$ V).

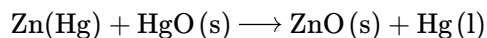
Batteries and fuel cells — *four devices, and the one fact asked about each*

Primary — cannot be recharged

Dry cell (Leclanché): zinc container = anode, graphite rod surrounded by MnO_2 and carbon = cathode, moist $\text{NH}_4\text{Cl}/\text{ZnCl}_2$ paste = electrolyte. Used in transistors and clocks.

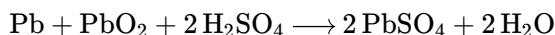


Mercury cell: used in hearing aids and watches. Its selling point — and the exam answer — is that **the potential stays constant through its life**, because no ions appear in the overall reaction, so no concentration changes:



Secondary — rechargeable

Lead storage battery (automobiles, inverters): Pb anode, PbO₂ on a lead grid as cathode, 38% H₂SO₄ electrolyte.

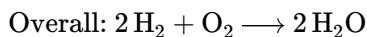
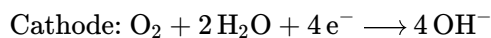


On **charging** the whole thing runs backwards: $2\text{PbSO}_4 + 2\text{H}_2\text{O} \longrightarrow \text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4$.

Nickel-cadmium: Cd anode, NiO₂ cathode, KOH electrolyte. Longer life than lead storage but more expensive to make.

Fuel cell

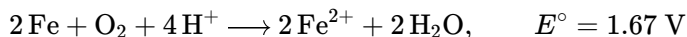
Converts a fuel's chemical energy *directly* to electricity, with reactants fed in continuously. The H₂-O₂ cell powered the **Apollo programme**, and its product water was condensed into the astronauts' drinking supply.



Two advantages (the standard 2-marker): ~70% efficiency against ~40% for a thermal plant, and it's pollution-free.

Corrosion — the electrochemical explanation, and two preventions

Corrosion is chemical attack by atmospheric gases and moisture on a metal surface, giving oxides, sulphides and carbonates. Rusting is the electrochemical case:



Fe²⁺ is then oxidised further by atmospheric oxygen to Fe³⁺, and rust is **hydrated ferric oxide**, Fe₂O₃ · xH₂O.

Prevention

- **Barrier protection** — paint, bisphenol, or a coat of another metal (Sn, Zn) between iron and the air.

- **Sacrificial protection** — attach a *more reactive* metal (Mg, Zn). It has the more negative E° , so it oxidises in preference to the iron and corrodes away while the iron survives.

Examiner asks: "why are magnesium blocks fixed to iron pipelines" — sacrificial anode; magnesium is more reactive, more negative E° , oxidises preferentially.

Numerical patterns, collected — *five patterns, one model each*

A · Nernst / EMF — 3 marks

Recognise it: concentrations given that aren't 1 M.

1. Write both half-reactions; balance electrons to get n .
2. $E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$.
3. Apply $E = E^\circ - \frac{0.0591}{n} \log Q$, coefficients as powers, solids omitted.

B · ΔG° and K_c — 3 marks

Recognise it: F is given, or the question says "at equilibrium".

1. Get E_{cell}° and n .
2. $\Delta_r G^\circ = -nFE^\circ$; divide by 1000 for kJ.
3. For K_c : $E^\circ = \frac{0.0591}{n} \log K_c$.

C · Conductivity chain — 3-5 marks

Recognise it: a resistance and either a cell constant or a known- κ reference solution.

1. Cell constant: $G^* = \kappa \times R$ from the reference solution, or read it off directly.
2. $\kappa = G^*/R$ for the unknown.
3. $\Lambda_m = \kappa \times 1000/M$.
4. If asked for α : divide by Λ_m° from Kohlrausch.

D · Faraday first law — 3 marks

Recognise it: current in amperes and a time.

1. Time $\rightarrow$ seconds.
2. Find n from the ion's charge.
3. $w = Mit/nF$, rearranged for whatever's missing.

E · Faraday second law (cells in series) — 5 marks

Recognise it: two electrolytic cells "connected in series".

1. Series means identical charge through both.
2. Use the first law on the cell you have full data for, to get time.
3. Then $w_1/w_2 = E_1/E_2$ with E = atomic mass/ n .

Worked · 2020 — two cells in series

2 A through ZnSO_4 (cell A) and CuSO_4 (cell B) in series; 2 g Cu deposited. $M(\text{Cu}) = 63.5$, $M(\text{Zn}) = 65$.

Time: $2 = \frac{63.5 \times 2 \times t}{2 \times 96500} \Rightarrow t = \frac{2 \times 96500}{63.5} = 3039 \text{ s}$

Zinc: both gain 2 electrons, so

$$\frac{w_{\text{Zn}}}{2} = \frac{65/2}{63.5/2} \Rightarrow w_{\text{Zn}} = \frac{65 \times 2}{63.5} = 2.04 \text{ g}$$

Past year questions · question types — *ranked by how often they turn up*

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · Nernst / EMF calculation — 3 marks

Recognise it: a cell notation with concentrations attached.

1. Balance for n .
2. E°_{cell} from the two E° values.
3. Nernst, watching the powers in Q .

Trap: sign error when log of a negative power comes down. $-\frac{0.0591}{n} \log 10^{-3}$ becomes *plus*.

2 · Conductivity / molar conductivity chain — 3-5 marks

Recognise it: a resistance value in ohms.

1. Cell constant.
2. κ , then ρ if asked.
3. Λ_m , then α if asked.

Trap: forgetting the $\times 1000$ in Λ_m , or dividing by the wrong concentration.

3 · Faraday's laws numerical — 3-5 marks

Recognise it: amperes, minutes, or "how many faradays".

1. Seconds.
2. n from the ion.
3. $w = Mit/nF$, or reason directly in moles of electrons.

4 · Electrolysis products with reasoning — 2-3 marks

Recognise it: "predict the products", with E° values supplied.

1. Cathode: higher E° reduced.
2. Anode: lower E° oxidised.
3. Check for the overpotential exception before answering.

5 · Cell/battery definitions and comparisons — 2 marks

Recognise it: "define", "name the cell used in...", "two differences".

1. Name the device.
2. Give both electrode reactions if asked.
3. For differences, answer in matched pairs.

6 · Kohlrausch application — 2-3 marks

Recognise it: limiting molar conductivities of ions given.

1. Write the electrolyte as its ions with correct multipliers.
2. Sum or subtract as needed.
3. For weak electrolytes, build from three strong ones.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
Nernst / EMF	3-marker	5 min
Conductivity chain	3-marker, or part of a 5	5–7 min
Faraday's laws	3-marker; series cells is a 5	4–7 min
Electrolysis products	2–3 marker	3 min
Definitions / cell names	2-marker	2 min
Kohlrausch	2–3 marker	3 min
ΔG° / K_c	3-marker	4 min

The video splits these as short-answer type I (2 marks), type II (3 marks), and long answer (5 marks) — the conductivity chain and series-cell problems are the ones that appear as 5-markers.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2019 Q6 · 2019 Q8 · lecture PYQ

Electrolysis of aqueous NaCl, and the overpotential reason chlorine appears instead of oxygen. Runs almost every year in some form.

2020 Q16 · lecture PYQ

Two electrolytic cells in series, ZnSO_4 and CuSO_4 , 2 g copper deposited — identical numbers both times.

2024 Q18 · 2026 Q22 · 2026 Q24

The conductivity → resistivity → molar conductivity chain, sometimes with degree of dissociation added. The single most reliable numerical in the chapter.

2025 Q1 · 2026 Q7(a)

Fuel cells: definition and two advantages, then the Apollo hydrogen-oxygen cell and its drinking-water by-product.

2022 Q13 · 2020 Q21 · 2026 Q22

Kohlrausch's law statement, then applied — degree of dissociation of acetic acid, limiting ionic conductivity of Al^{3+} , Λ° of NaCl.

2020 Q3 · 2026 Q23(b)

Electrochemical vs electrolytic cell differences, and primary vs secondary battery differences. Straight recall, always available.

Which cell is used where — dry cell in transistors, mercury cell in hearing aids and watches, with electrode reactions.

Past year questions · cold practice — *answers only* — *work them before you look*

Nernst / EMF

- 2020 Q2 — write cell notation and the Nernst equation for $\text{Mg} + 2 \text{Ag}^+ \longrightarrow \text{Mg}^{2+} + 2 \text{Ag}$.
 $\text{Mg}|\text{Mg}^{2+}||\text{Ag}^+|\text{Ag}; E = E^\circ - \frac{0.0591}{2} \log \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$
- 2026 Q19 — EMF of $\text{Sn}|\text{Sn}^{2+}||\text{H}^+|\text{H}_2$ at 298 K with concentrations given. Apply Nernst with $n = 2$; H_2 omitted as a gas at 1 bar

ΔG° and K_c

- 2026 Q23 — E° for $\text{Cu} + 2 \text{Ag}^+ \rightleftharpoons \text{Cu}^{2+} + 2 \text{Ag}$ at equilibrium, $K_c = 10^{15}$. $E^\circ = \frac{0.0591}{2} \times 15 = 0.44$ V

Conductivity

- Lecture PYQ — 0.05 M KCl, $l = 50$ cm, $A = 0.625$ cm², $R = 5 \times 10^3$ Ω . Find ρ , κ , Λ_m . 62.5 Ω cm; 0.016 S cm⁻¹; 320 S cm² mol⁻¹
- Lecture PYQ — cell gives 164 Ω with 0.02 M KCl ($\kappa = 2.768 \times 10^{-3}$), then 78.5 Ω with 0.05 M AgNO₃. Find κ and Λ_m of AgNO₃. $G^* = 0.4539$; $\kappa = 5.78 \times 10^{-3}$; $\Lambda_m = 115.6$ S cm² mol⁻¹
- 2026 Q22 — 0.1 M NaCl, $\kappa = 1.06 \times 10^{-2}$ S cm⁻¹, $\lambda^\circ \text{Na}^+ = 50.1$, $\text{Cl}^- = 76.5$. Find Λ_m and α . 106 S cm² mol⁻¹; $\Lambda^\circ = 126.6$; $\alpha = 0.837$
- 2022 Q13(b) — α of acetic acid if $\Lambda_m = 48$ and $\Lambda^\circ_m = 400$. 0.12

Kohlrausch

- 2021-22 — Λ° of MgCl₂ given $\lambda^\circ \text{Mg}^{2+} = 106$, $\text{Cl}^- = 76.3$. 285.6 S cm² mol⁻¹ (remember the 2 $\times$)

Faraday's laws

- 2014 — 5 A through Ni(NO₃)₂ for 20 min, $M = 58.7$. Mass of Ni? 1.825 g
- 2015 — charge to reduce 1 mol Zn²⁺ to Zn. 2 F = 193000 C
- 2026 Q25(b) — faradays to produce 40 g Al from molten Al₂O₃, $M = 27$. 4.44 F

Built from Sourabh Raina's Electrochemistry one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 2 (Rationalised 2022-23). Verified against NCERT:

"Kohlrausch law of independent migration of ions"; Faraday constant 96487 C mol⁻¹ (taken as 96500 in working); Daniell cell 1.1 V at unit concentration.